

Barangay Relief Pack Optimizer

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1. Application Overview

The Barangay Relief Pack Optimizer is a desktop application built in Python using Tkinter. It is designed to help barangay disaster response officers make two critical decisions quickly and accurately after a typhoon: what relief goods to pack into each bag, and which families should receive bags first when supply is limited.

The application has three screens accessible from the left sidebar:

Inventory Setup - enter relief goods and run the bag optimizer

Family Registry - register affected families and generate the assignment

Results - view the optimized bag and ranked assignment report

2. Screen 1: Inventory Setup

2.1 Purpose

This screen is where the officer enters the available relief goods, sets the maximum bag weight, and specifies how many bags are available. Clicking Optimize Bag runs the 0/1 Knapsack DP algorithm and produces the optimal bag composition.

The screenshot shows the 'Inventory Setup' screen of the 'Relief Pack Optimizer' application. The interface includes a dark blue sidebar on the left with the application title and navigation links: 'Barangay System', 'Inventory Setup' (highlighted), 'Family Registry', 'Results', and a red 'New Disaster' button. Below these are input fields for 'Disaster:' and 'Barangay:'. The main content area is titled 'Inventory Setup' and contains a subtitle 'Enter available relief goods, set bag capacity and bag count.'. It features two input fields at the top: 'Max bag weight (kg):' set to 10.0 and 'Bags available:' set to 0. Below these is a table titled 'Relief Items' with columns for 'Item name', 'Weight (kg)', 'Benefit (1-10)', 'Qty', and 'Action'. The table contains six rows of items: Rice, Canned goods, Water, Medicine, Blanket, and Hygiene kit. Each row has input fields for the weight, benefit, and quantity, and a red trash icon in the action column. A '+ Add row' button is located at the top right of the table. At the bottom right of the main area is a green 'Optimize Bag' button.

Item name	Weight (kg)	Benefit (1-10)	Qty	Action
Rice	3.0	9	50	
Canned goods	1.5	7	80	
Water	2.0	8	60	
Medicine	0.5	10	40	
Blanket	1.5	5	30	
Hygiene kit	0.5	6	50	

2.2 Input Fields

Max bag weight (kg): The maximum weight a single relief bag can hold. Entered using a numeric spinner.

Bags available: The total number of relief bags the barangay can distribute.

Relief Items Table: Each row represents one item type with four fields: Item name, Weight (kg), Benefit score (1–10), and Quantity available.

Default items pre-loaded into the table:

- Rice - 3.0 kg, benefit 9
- Canned goods - 1.5 kg, benefit 7
- Water - 2.0 kg, benefit 8
- Medicine - 0.5 kg, benefit 10
- Blanket - 1.5 kg, benefit 5
- Hygiene kit - 0.5 kg, benefit 6

2.3 How to Use

1. Set the maximum bag weight using the spinner (e.g., 10 kg).
2. Set the number of bags available.
3. Review or edit the relief items table. Use + Add row to add new items. Click the trash icon to remove a row.
4. Click Optimize Bag. The system runs the DP algorithm and displays a confirmation showing the number of items selected, total weight, and total benefit score.

The screenshot shows the 'Relief Pack Optimizer' application window. On the left is a dark blue sidebar with navigation links: 'Barangay System', 'Inventory Setup' (highlighted), 'Family Registry', 'Results', and a red 'New Disaster' button. Below these are input fields for 'Disaster:' and 'Barangay:'. The main area is titled 'Inventory Setup' with the instruction 'Enter available relief goods, set bag capacity and bag count.' It features two spinners: 'Max bag weight (kg):' set to 10.0 and 'Bags available:' set to 5. Below is a table titled 'Relief Items' with columns: Item name, Weight (kg), Benefit (1–10), Qty, and Action. The table contains six rows of default items. A '+ Add row' button is at the top right of the table. A modal dialog box is open in the center, titled 'Done!', with an information icon. It displays: 'Optimal bag packed!', 'Items: 6', 'Weight: 9.0 / 10.0 kg', 'Benefit: 45', and 'Go to Family Registry next.' with an 'OK' button. At the bottom left of the main area, a green status bar says 'Bag optimized: 6 items, 9.0 kg, benefit=45'. At the bottom right is a green 'Optimize Bag' button.

Item name	Weight (kg)	Benefit (1–10)	Qty	Action
Rice	3.0	9	50	🗑️
Canned goods	1.5	7	80	🗑️
Water	2.0	8	60	🗑️
Medicine	0.5	10	40	🗑️
Blanket	1.5	5	30	🗑️
Hygiene kit	0.5	6	50	🗑️

2.4 What Happens Internally

The Optimize Bag button triggers the 0/1 Knapsack DP algorithm in knapsack.py. Item weights are scaled to 0.1 kg precision to allow integer DP table indexing. A 2D table $dp[i][w]$ is filled bottom-up where each cell stores the maximum benefit achievable using the first i items with capacity w . The algorithm then backtracks through the table to identify exactly which items were selected.

3. Screen 2: Family Registry

3.1 Purpose

This screen is where the officer registers each affected family. Each family is automatically assigned a priority score based on the formula below. Clicking Generate Assignment runs the greedy ranking and bag assignment.

Relief Pack Optimizer
Barangay System

Inventory Setup

Family Registry

Results

New Disaster

Disaster:

Barangay:

Family Registry
Register affected families. Priority scores are computed automatically.

Family ID: F-001 Size: Vulnerable members: 0 Damage level (1-3): 1 **Add Family**

Families: 0 **Clear all**

Family ID	Size	Vulnerable	Damage	Formula score	Action

Generate Assignment

3.2 Priority Score Formula

$$\text{priority_score} = (\text{family_size} \times 2) + (\text{vulnerable_count} \times 5) + (\text{damage_level} \times 3)$$

family_size: Total number of household members

vulnerable_count: Number of elderly, infants, or pregnant members (weighted highest at 5 - greatest health risk)

damage_level: 1 = minor damage, 2 = partial loss, 3 = total loss

Tiebreaker rule: if two families have equal scores, the larger household is ranked first. If still equal, the family registered earlier takes precedence.

3.3 Input Fields

Family ID: A unique identifier for the household (e.g., F-001)

Size: Total number of household members (1–20)

Vulnerable members: Count of elderly, infants, or pregnant members

Damage level: Selected from dropdown: 1 (minor), 2 (partial), 3 (total loss)

The screenshot shows the 'Relief Pack Optimizer' application window. On the left is a dark blue sidebar with navigation links: 'Relief Pack Optimizer', 'Barangay System', 'Inventory Setup', 'Family Registry' (highlighted), and 'Results'. Below these is a red 'New Disaster' button and input fields for 'Disaster:' and 'Barangay:'. The main content area is titled 'Family Registry' with a subtitle 'Register affected families. Priority scores are computed automatically.' It features a form with four input fields: 'Family ID' (containing 'F-005'), 'Size' (containing '6'), 'Vulnerable members' (containing '1'), and 'Damage level (1-3)' (a dropdown menu showing '3'). To the right of these fields is a blue 'Add Family' button and a green status message 'Added F-004 score=26'. Below the form is a table titled 'Families: 4' with a 'Clear all' button. The table has columns: 'Family ID', 'Size', 'Vulnerable', 'Damage', 'Formula score', and 'Action'. It lists four families: F-001 (Size 5, Vulnerable 2, Damage 1, Score 23), F-002 (Size 5, Vulnerable 2, Damage 2, Score 26), F-003 (Size 5, Vulnerable 1, Damage 2, Score 21), and F-004 (Size 6, Vulnerable 1, Damage 3, Score 26). Each row has a 'Remove' button. At the bottom right of the table area is a green 'Generate Assignment' button.

Family ID	Size	Vulnerable members	Damage level (1-3)
F-005	6	1	3

Families: 4 Clear all

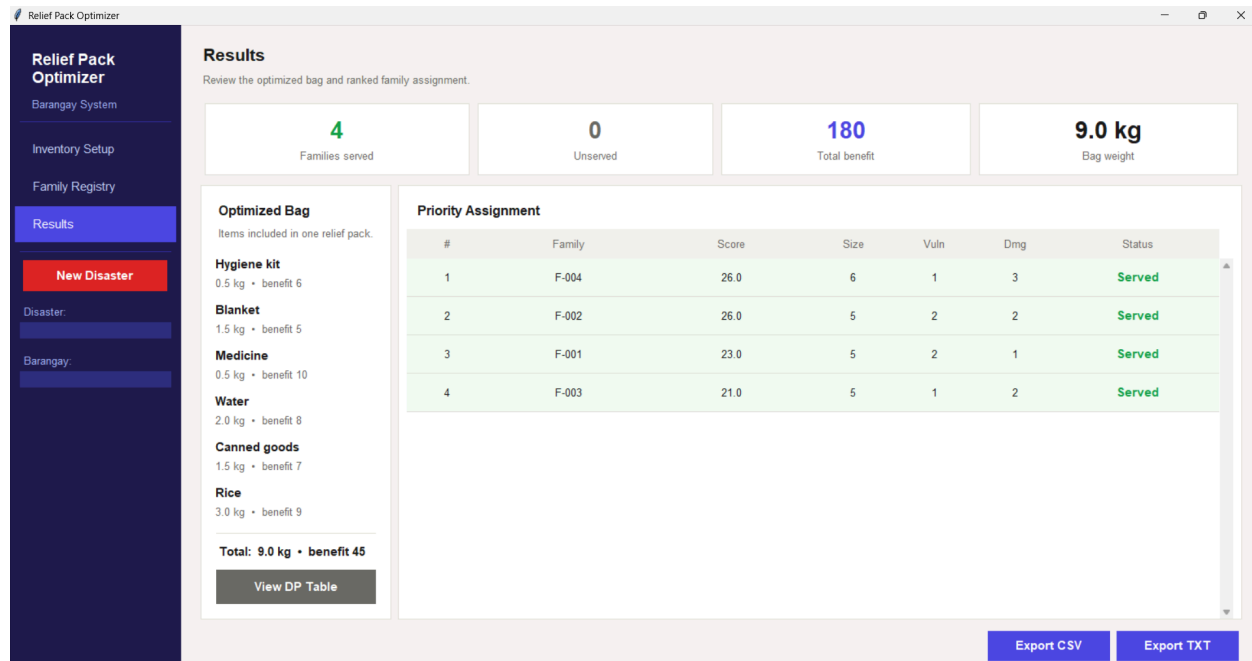
Family ID	Size	Vulnerable	Damage	Formula score	Action
F-001	5	2	1	23	Remove
F-002	5	2	2	26	Remove
F-003	5	1	2	21	Remove
F-004	6	1	3	26	Remove

Generate Assignment

4. Screen 3: Results

4.1 Purpose

This screen displays the complete output of both algorithms, the optimized bag manifest on the left and the ranked family assignment table on the right. A summary strip at the top shows key statistics at a glance.



4.2 Summary Strip

Four cards at the top of the screen display:

- Families served - how many families received a bag
- Unserved - how many families did not receive a bag due to supply shortage
- Total benefit - total benefit score delivered across all served families
- Bag weight - the total weight of one optimized bag

4.3 Optimized Bag Manifest (Left Panel)

Lists every item selected by the DP algorithm for one relief bag. Each item shows its name, weight, and benefit score. The total weight and total benefit are displayed at the bottom.

4.4 Priority Assignment Table (Right Panel)

Lists all registered families ranked by priority score from highest to lowest. Each row shows the family ID, final score, household size, vulnerable count, damage level, and status (Served or No Supply). Served families are highlighted in green, unserved in red.

4.5 View DP Table

The View DP Table button opens a separate window displaying the full 0/1 Knapsack DP table used in Module 1. Rows represent items added one by one. Columns represent bag capacity shown in 0.5 kg steps for readability. Internally the algorithm works at 0.1 kg precision, so the full table has more columns than displayed. Each cell shows the maximum benefit achievable at that item count and capacity. This is provided for transparency and academic verification of the algorithm.

Relief Pack Optimizer

Results

DP Table — 0/1 Knapsack

0/1 Knapsack DP Table (rows = items added, columns = capacity in 0.1 kg steps)

Item \ Cap	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0
Base	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Rice	0	0	0	0	0	0	9	9	9	9	9	9	9	9	9	9	9
Canned goods	0	0	0	7	7	7	9	9	9	16	16	16	16	16	16	16	16
Water	0	0	0	7	8	8	9	15	15	16	17	17	17	24	24	24	24
Medicine	0	10	10	10	17	18	18	19	25	25	26	27	27	27	34	34	34
Blanket	0	10	10	10	17	18	18	22	25	25	26	30	30	31	34	34	34
Hygiene kit	0	10	16	16	17	23	24	24	28	31	31	32	36	36	37	40	40

Total benefit
180

Size	Vuln
6	1
5	2
5	2
5	1

Canned goods
1.5 kg • benefit 7

Rice
3.0 kg • benefit 9

4.6 Export Options

Two export buttons are available at the bottom of the Results screen:

- Export TXT - generates a plain text report of the bag manifest and assignment list, suitable for printing and field use
- Export CSV - generates a spreadsheet-compatible file of the assignment table for record keeping

Relief Pack Optimizer

Save As

Shaina - Personal > Documents >

File name: relief_report

Save as type: CSV

Save Cancel

2.0 kg • benefit 8

Canned goods
1.5 kg • benefit 7

Rice
3.0 kg • benefit 9

Total: 9.0 kg • benefit 45

View DP Table

225
Total benefit

9.0 kg
Bag weight

Score	Size	Vuln	Dmg	Status
33.0	6	3	2	Served
31.0	5	3	2	Served
26.0	5	2	2	Served
18.0	5	1	1	Served
13.0	5	0	1	Served

Export CSV Export TXT

